

Discovery Executive Summary

Project: PingCRM-Discovery · Generated: 30/09/2026, 11:37:21

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Testing & Quality Assurance Analysis	—

1. Testing & Quality Assurance Analysis

EXECUTIVE SUMMARY

Ping CRM is a Laravel 11 + React 19 (Inertia.js) application with a large IVR enterprise module surface. The test suite is critically thin: only 3 meaningful PHPUnit test files cover 2 of 141 backend source files (2.1% file coverage), and the frontend has a single assertion-less smoke test for 904 TypeScript/React source files (0.1%). Authentication, user management, the entire IVR enterprise module (91 controllers, 12 legacy "GodService" classes, 12 repositories), reports, and all 14 shared UI components ship with zero automated test coverage. The CI pipeline (`tests.yml`) runs PHPUnit on push/PR but does not run Vitest, so the frontend is never gate-checked. No integration tests exercise real database boundaries beyond the two existing feature tests, no contract tests exist for the IVR module's dynamic routing, and no end-to-end tests verify user journeys.

5.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	Untested Critical Logic	Critical modules with zero tests	0	1–3	>3	7 critical modules (auth, users, reports, IVR hub, IVR module controllers, legacy services, repositories)	High Risk
H2	Low Test Coverage	Overall coverage %	>80%	50–80%	<50%	~2% backend / ~0% frontend (estimated from test-file-to-source ratio)	High Risk
H3	Missing Integration Tests	Boundaries covered %	>70%	30–70%	<30%	~5% — only ContactsTest and OrganizationsTest exercise DB	High Risk
H4	Missing Contract Tests	APIs with contract tests %	>80%	40–80%	<40%	0% — no contract or schema validation tests exist	High Risk
H5	Flaky / Skipped Tests	Skipped/flaky test count	0	1–5	>5	0	Good
H6	No CI Test Gate	Tests enforced in CI	Required gate	Runs, not required	No CI test run	PHPUnit runs on push/PR (not required); Vitest never runs in CI	High Risk
H7	No End-to-End Tests (additional)	E2E test suites present	≥1 suite	Partial/smoke only	None	0 — no Cypress, Playwright, or Dusk configured	High Risk
H8	Assertion-Free Tests (additional)	Tests with no meaningful assertions	0	1–2	>2	2 — <code>ExampleTest.php</code> and <code>smoke.test.ts</code>	Moderate

5.4 Actions Required

Hotspot	Action	Rating	Priority
H1 — Untested Critical Logic	Generate unit and feature tests for auth, users, reports, IVR controllers, legacy services, and repositories	High Risk	Critical
H2 — Low Test Coverage	Enable coverage reporting; target 50% backend and 30% frontend coverage in first sprint; set CI thresholds	High Risk	Critical

H3 — Missing Integration Tests	Add integration tests for ReportsController raw queries, IVR Store/Update/Destroy cycles, and auth session flow	High Risk	High
H4 — Missing Contract Tests	Add Inertia assertInertia tests for all routes; add CSV output format tests; parameterized test for IVR SLUG_MAP	High Risk	High
H6 — No CI Test Gate	Add Vitest to CI pipeline; configure branch protection requiring tests + static-analysis to pass	High Risk	High
H7 — No End-to-End Tests	Install Playwright or Dusk; write smoke E2E for login, IVR navigation, and CRUD flows; add to CI	High Risk	High
H8 — Assertion-Free Tests	Replace ExampleTest.php and smoke.test.ts with real application assertions	Moderate	Medium

5.5 Expected Outcomes

- **Critical auth and user-management paths protected** — login, logout, rate limiting, user CRUD, and demo-user guards verified automatically before every merge, preventing security regressions.
- **IVR enterprise module regression-safe** — the 91 IVR controllers and 12 legacy services gain integration test coverage, catching broken DB queries and data corruption before production.
- **CI catches regressions on both layers** — with Vitest added to the pipeline and branch protection enabled, no code (backend or frontend) merges without passing tests.
- **Contract tests prevent silent Inertia breaks** — every route's prop structure is validated, eliminating the class of bugs where PHP changes break React rendering at runtime.
- **E2E tests verify real user journeys** — the login-to-IVR-module flow and CRUD operations are tested in a real browser, catching hydration failures and navigation bugs that unit tests cannot detect.

```

{
  "stop_reason": "end_turn",
  "session_id": "6a91ab74-397f-4455-adbf-213b2f5f78e9",
  "total_cost_usd": 1.9992385000000001,
  "usage": {
    "input_tokens": 21,
    "cache_creation_input_tokens": 84591,
    "cache_read_input_tokens": 1148683,
    "output_tokens": 22883,
    "server_tool_use": {
      "web_search_requests": 0,
      "web_fetch_requests": 0,
      "service_tier": "standard",
      "cache_creation": {
        "ephemeral_1h_input_tokens": 84591,
        "ephemeral_5m_input_tokens": 0,
        "inference_geo": "not_available",
        "iterations": [
          [
            {
              "input_tokens": 1,
              "output_tokens": 1713,
              "cache_read_input_tokens": 92916,
              "cache_creation_input_tokens": 481,
              "cache_creation": {
                "ephemeral_5m_input_tokens": 0,
                "ephemeral_1h_input_tokens": 481,
                "type": "message"
              }
            },
            {
              "speed": "standard",
              "modelUsage": {
                "claude-haiku-4-5-20251001": {
                  "inputTokens": 6727,
                  "outputTokens": 16,
                  "cacheReadInputTokens": 0,
                  "cacheCreationInputTokens": 0,
                  "webSearchRequests": 0,
                  "costUSD": 0.0068070000000000006,
                  "contextWindow": 200000,
                  "maxOutputOpus-4-6": {
                    "inputTokens": 21,
                    "outputTokens": 22883,
                    "cacheReadInputTokens": 1148683,
                    "cacheCreationInputTokens": 84591,
                    "webSearchRequests": 0,
                    "costUSD": 1.9924315000000001,
                    "contextWindow": 200000,
                    "terminal_reason": "completed",
                    "fast_mode_state": "off",
                    "uuid": "ae43520a-e665-467d-a389-1d2dff37e858"
                  }
                }
              }
            }
          ]
        }
      }
    }
  }
}

```